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FOOTPRINTING &
RECONNIACCENCE
MODULE 2

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INTRODUCTION TO FOOTPRINTING & RECONNAISSANCE

Introduction to Footprinting & Reconnaissance

FOOTPRINTING CONCEPTS

This step acts as a preparatory phase for the attacker, who needs to gather as much information as possible to easily find ways to intrude the target network

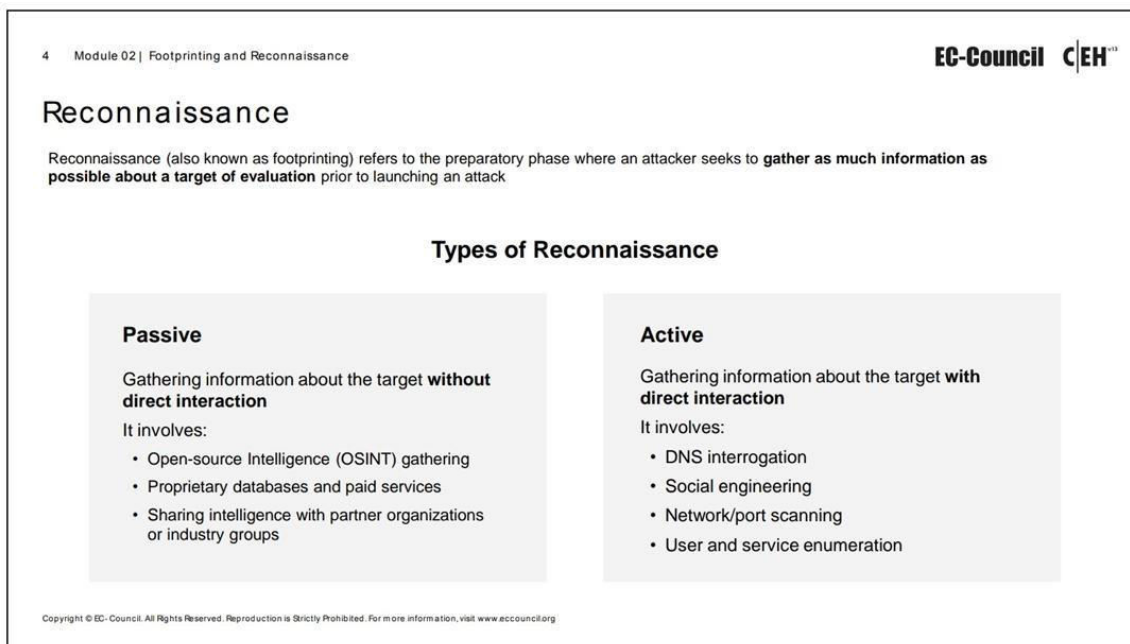
RECOINNAISSEANCE

Reconnaissance (also known as Footprinting) refers to the preparatory phase where an attacker seeks to gather as much information as possible about a target of evaluation prior to launching an attacker. An essential aspect of footprint is identifying the level of risk associated with the organization's publicly accessible information. Footprinting, the first step in ethical hacking, refers to the process of collecting information about a target network and its environment. Using Footprinting, you can find a number of opportunities to penetrate and assess the target organization's network.

Key Concepts:

1. **Attack surface:** exposed resources (ports, services, apps).
2. **Passive vs Active:** passive gathers without direct interaction, active involves direct queries/scans.
3. **OSINT:** Open-Source Intelligence — a large component of passive Footprinting.

Fig. Types of Reconnaissance

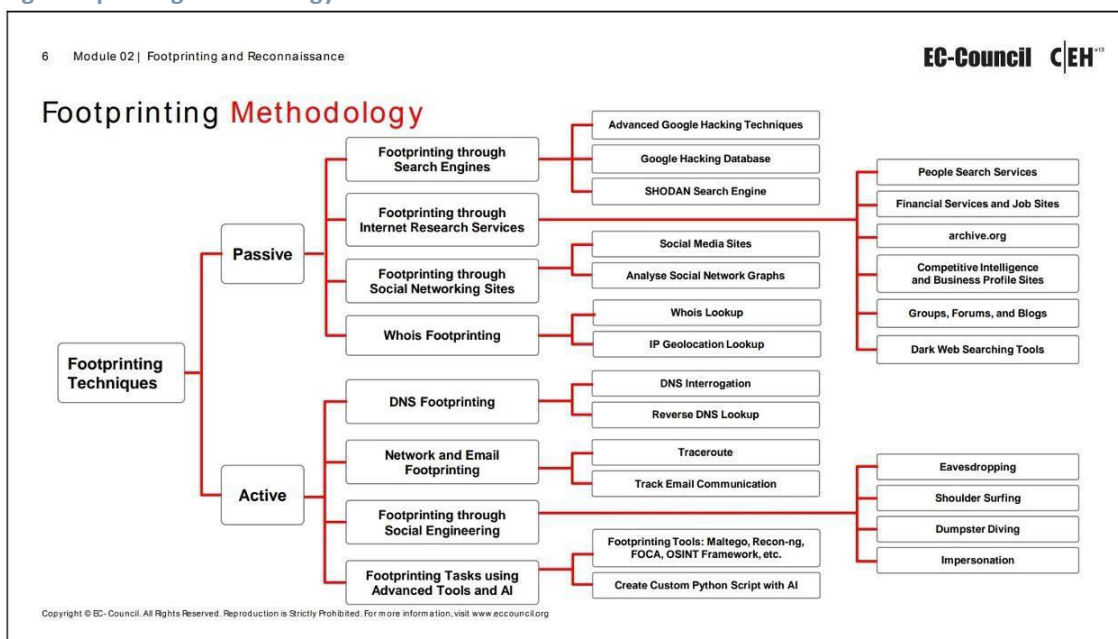


INTRODUCTION TO FOOTPRINTING & RECONNAISSANCE

FOOTPRINTING METHODOLOGY

Footprinting methodology uses passive and active techniques to build a comprehensive profile of a target's digital and physical presence. Passive methods include search engines, WHOIS, social media analysis, archives and OSINT; active methods involve DNS interrogation, port scanning, traceroute and targeted social engineering. Practitioners use tools like Maltego, Shodan, and recon frameworks to identify attack surface and vulnerabilities.

Fig. Footprinting Methodology



TYPES OF FOOTPRINTING/RECONNAISSANCE

Footprinting can be categorized into Passive Footprinting and Active Footprinting:

- **Passive Footprinting:** It involves Gathering information about the target without direct interaction. It is mainly useful when information gathering activities are not to be detected by the target. It involves
 - **Open-Source Intelligence (OSINT) Gathering.**
 - **Proprietary databases and paid services.**
 - **Sharing Intelligence with partner organizations or industry groups.**
- **Active Footprinting:** It involves gathering information about the target with direct interaction. In active Footprinting target may recognize the ongoing information gathering process, as we overtly interact with the target network. It involves:
 - **DNS interrogation**
 - **Social Engineering**

INTRODUCTION TO FOOTPRINTING & RECONNAISSANCE

- **Network/Port Scanning**
- **User and service enumeration**

OBJECTIVE OF FOOTPRINTING & RECONNAISSANCE

The objective of Footprinting and Reconnaissance is to systematically collect and analyze information related to the target organization's network architecture, digital assets, and publicly accessible resources. This phase focuses on obtaining accurate and comprehensive intelligence—including domain details, IP addressing, DNS configurations, service information, employee data, and associated third-party dependencies—using a combination of passive and active information-gathering methodologies. The goal is to establish a clear understanding of the target's external security posture, identify potential exposure points, and provide a well-informed foundation for subsequent penetration testing activities, while maintaining adherence to ethical and legal guidelines.

1. GOOGLE DORKING

1. Google Dorking

Information gathering is the first and one of the most critical phases in a penetration test. Advanced Google Hacking techniques, also known as **Google Dorking**, utilize Google's powerful search operators to discover publicly exposed sensitive information, misconfigurations, and security vulnerabilities. This phase helps assess an organization's digital footprint and identify risks arising from unintended data exposure.

1.1 OBJECTIVE:

- Identify publicly accessible sensitive information.
- Enumerate exposed directories, files, and services.
- Detect misconfigurations such as open indexes or leaked credentials.
- Understand how much information about the target is available online.
- Assist in mapping potential attack surfaces for further testing.

1.2 SCOPE:

This assessment includes publicly available data indexed by Google. No intrusive scanning or active exploitation is performed in this stage.

The scope covers:

- Domain-based searches
- Filetype searches
- Sensitive exposure checks
- Misconfiguration detection
- Public footprint analysis

Target: <https://google-gruyere.appspot.com/> A purposely vulnerable web application created by Google Security team.

1.3 METHODOLOGY:

During information gathering, the following Google operators and dorking techniques were used:

Basic Operators:

- **site:** Restricts results to a specific domain.
- **intitle:** Searches for pages with specific text in the title.
- **inurl:** Searches for keywords within the URL.
- **filetype:** Finds disclosed files (.pdf, .xls, .txt, .log, etc.).
- **cache:** Views cached versions of pages.

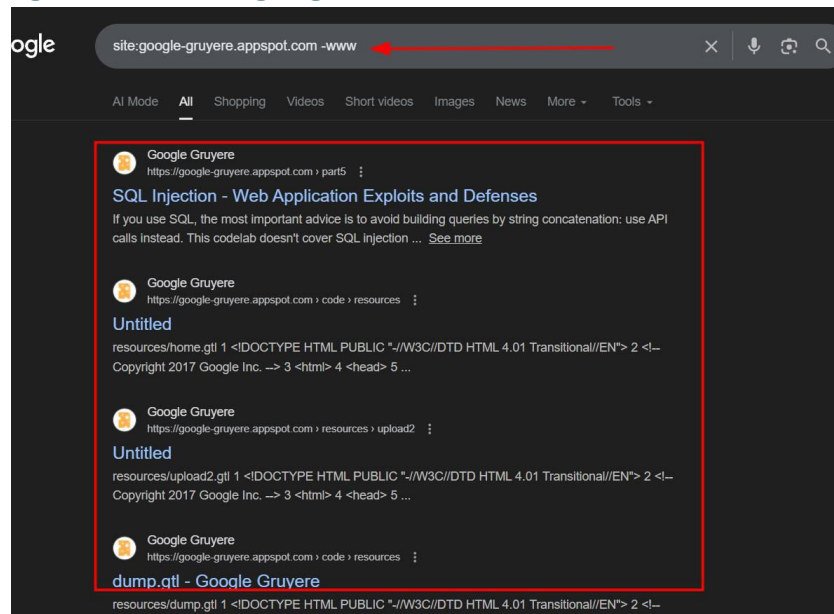
1. GOOGLE DORKING

1. Basic site targeting

site:google-gruyere.appspot.com

site:google-gruyere.appspot.com -www

Fig. 1.3.1 Basic Site targeting



Advanced Google Dorks:

- Searching for exposed admin portals
- Identifying directory listings
- Finding exposed configuration files (.env, .conf, .ini)
- Searching for leaked credentials or database dumps
- Locating sensitive documents such as invoices or internal reports
- Finding login pages using fingerprinting keywords

2. Directory & File Enumeration Dorks

site:google-gruyere.appspot.com intitle:index.of

site:google-gruyere.appspot.com "Directory listing"

site:google-gruyere.appspot.com inurl:/examples/

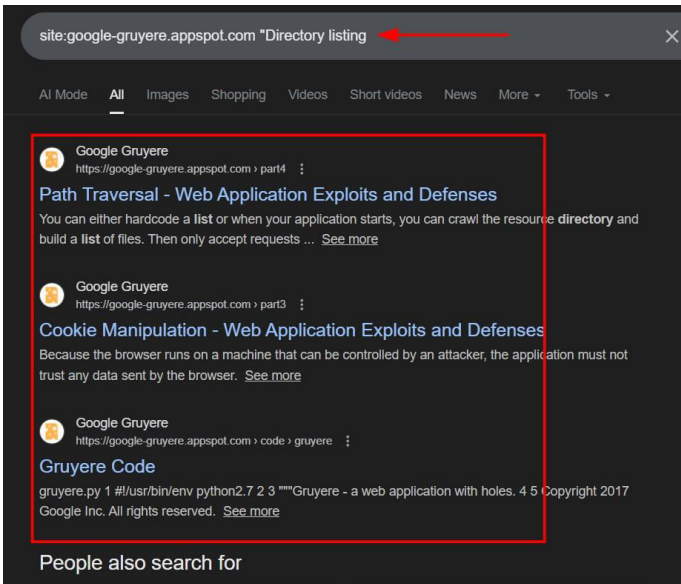
site:google-gruyere.appspot.com inurl:/snippets/

site:google-gruyere.appspot.com inurl:/uploads/

site:google-gruyere.appspot.com inurl:feed.gtl

1. GOOGLE DORKING

fig. 1.3.2 Directory enumeration Dorks



3. Parameter Discovery (Query String Enumeration)

Gruyere uses parameters like `?id=`, `?snip=`, `?hacker=` etc.

`site:google-gruyere.appspot.com inurl:id=`

`site:google-gruyere.appspot.com inurl:snip=`

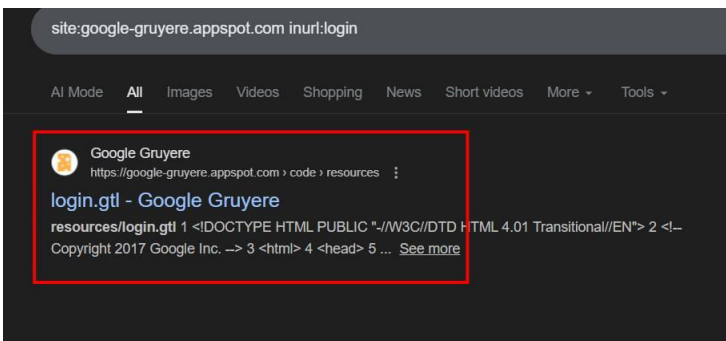
`site:google-gruyere.appspot.com inurl:hacker=`

`site:google-gruyere.appspot.com inurl:profile=`

`site:google-gruyere.appspot.com inurl:edit=`

`site:google-gruyere.appspot.com inurl:loginfig.`

Fig. 1.3.3 Parameter Discovery



1. GOOGLE DORKING

4. Detect Publicly Exposed Snippets (Pages created by users)

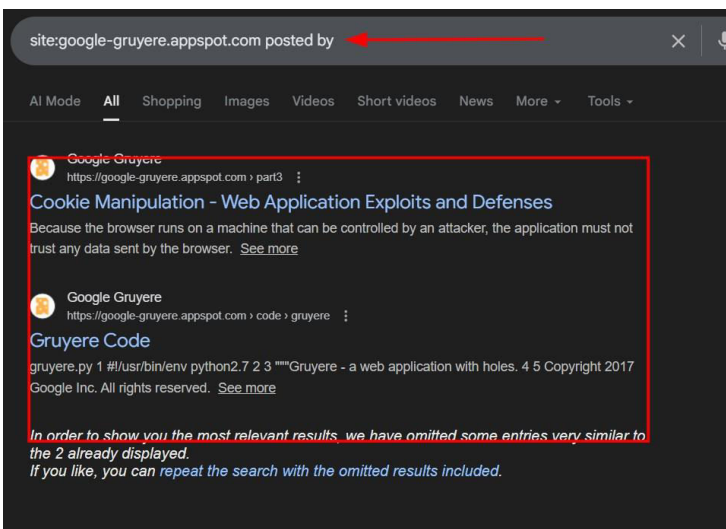
site:google-gruyere.appspot.com "posted by"

site:google-gruyere.appspot.com "snippet"

site:google-gruyere.appspot.com "New Snippet"

site:google-gruyere.appspot.com "New Profile"

fig. 1.3.4 Detect Publicly exposed Snippets



5. Filetype-Based Discovery

Gruyere contains **.py**, **.html**, **.gtl**, and **resource files**.

site:google-gruyere.appspot.com filetype:py

site:google-gruyere.appspot.com filetype:gtl

site:google-gruyere.appspot.com filetype:html

site:google-gruyere.appspot.com filetype:css

site:google-gruyere.appspot.com filetype:js

Source Code Exposure

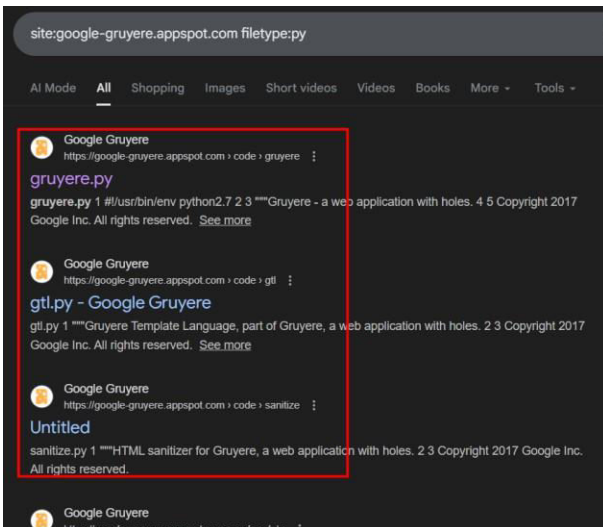
site:google-gruyere.appspot.com "def "

site:google-gruyere.appspot.com "import "

site:google-gruyere.appspot.com "class "

1. GOOGLE DORKING

fig. 1.3.5 Source Code Exposure



gruyere.py

```
1 #!/usr/bin/env python2.7
2
3 """Gruyere - a web application with holes.
4
5 Copyright 2017 Google Inc. All rights reserved.
6
7 This code is licensed under the
8 https://creativecommons.org/licenses/by-nd/3.0/us/
9 Creative Commons Attribution-No Derivative Works 3.0 United States license.
10
11 DO NOT COPY THIS CODE!
12
13 This application is a small self-contained web application with numerous
14 security holes. It is provided for use with the Web Application Exploits and
15 Defenses codelab. You may modify the code for your own use while doing the
16 codelab but you may not distribute the modified code. Brief excerpts of this
17 code may be used for educational or instructional purposes provided this
18 notice is kept intact. By using Gruyere you agree to the Terms of Service
19 https://www.google.com/intl/en/policies/terms/
20 """
21
22 __author__ = 'Bruce Leban'
23
24 # system modules
25 from BaseHTTPServer import BaseHTTPRequestHandler
26 from BaseHTTPServer import HTTPServer
27 import cgi
28 import cPickle
29 import os
30 import random
31 import sys
32 import threading
33 import urllib
34 from urlparse import urlparse
35
36 try:
37     sys.dont_write_bytecode = True
38 except AttributeError:
39     pass
40
41 # our modules
42 import data
43 import gtl
44
45 DB_FILE = '/stored-data.txt'
46 SECRET_FILE = '/secret.txt'
47
48 ...
```

6. Search for Hidden Pages

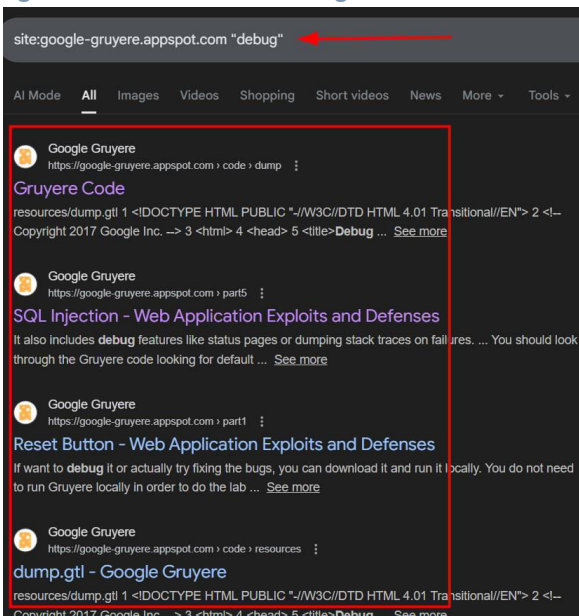
These often appear in labs:

site:google-gruyere.appspot.com "hidden"

site:google-gruyere.appspot.com "debug"

site:google-gruyere.appspot.com intitle:admin

fig. 1.3.6 Search For Hidden Pages



1. GOOGLE DORKING

1.4 FINDINGS SUMMARY

The gathered information includes:

- **Publicly exposed files or directories.**
- **Metadata leaks** from documents.
- **Technology stack information** revealed from page titles or cached pages.
- **Potential sensitive information exposure** like backup files, logs, or configuration data.

1.5 IMPACT

Exposed information may lead to:

- Unauthorized access to systems or admin portals
- Credential leaks
- Reconnaissance by attackers for targeted exploitation
- Sensitive data disclosure (customer data, internal documents)
- Increased attack surface for future penetration attempts

1.6 MITIGATIONS

To mitigate risks:

- Use **robots.txt** and **.htaccess** restrictions appropriately
- Restrict access to sensitive directories and internal portals
- Disable directory listing on web servers
- Remove exposed backup or log files
- Ensure sensitive documents are not publicly accessible
- Regularly monitor Google search results for unwanted exposure
- Implement a **Google Search Console** visibility report

1.7 CONCLUSION

Google Hacking is a highly effective, non-intrusive method of gathering intelligence about an organization. The data gathered in this phase helps security teams understand publicly available information and fix potential leak points before malicious actors exploit them.

2. WHOIS LOOKUP

2. Whois Lookup

WHOIS lookup is a passive information-gathering technique used during penetration testing to obtain publicly available registration details about a domain. For the target **testfire.net**, WHOIS information helps reveal administrative contacts, name servers, registrar details, and domain metadata. This information provides insight into the domain's ownership and can support further reconnaissance.

2.1 OBJECTIVE

The objective of performing WHOIS lookup on **testfire.net** is to:

- Identify domain registrar and registration details
- Enumerate DNS and name server information
- Discover administrative/technical contact information (if publicly available)
- Determine domain creation, expiration, and update history
- Understand hosting and infrastructure used by the domain
- Support overall reconnaissance for security assessment

2.2 SCOPE

This analysis is limited to publicly accessible data from WHOIS databases.

No intrusive or active exploitation is performed.

Scope includes:

- Domain registration data
- Registrar details
- Administrative and technical contacts
- DNS and name server information
- Domain lifecycle timeline (creation/expiry)

2.3 METHODOLOGY

The following steps were used to gather WHOIS information:

Tools Used:

- **whois** command-line utility
- Online **WHOIS lookup** services (e.g. <https://www.whois.com/>)

2. WHOIS LOOKUP

fig. 2.3.1 Results for WHIIS Commandline and Online Service

```
sai@kali: ~  
File Actions Edit View Help  
[sai@kali]~  
$ whois testfire.net  
Domain Name: TESTFIRE.NET  
Registry Domain ID: 8363973_DOMAIN_NET-VRSN  
Registrar WHOIS Server: whois.registrar.amazon  
Registrar URL: http://registrar.amazon.com  
Updated Date: 2025-02-27T17:53:32Z  
Creation Date: 1999-07-23T13:52:22Z  
Registry Expiry Date: 2026-07-23T13:52:32Z  
Registrar: Amazon Registrar, Inc.  
Registrar IANA ID: 468  
Registrar Abuse Contact Email: trustandsafety@support.aws.com  
Registrar Abuse Contact Phone: +1.2024422253  
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited  
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited  
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited  
Name Server: ASIA2.AKAM.NET  
Name Server: EUR2.AKAM.NET  
Name Server: EUR5.AKAM.NET  
Name Server: NS1-206.AKAM.NET  
Name Server: NS1-99.AKAM.NET  
Name Server: USC2.AKAM.NET  
Name Server: USC3.AKAM.NET  
Name Server: USW2.AKAM.NET  
DNSSEC: unsigned  
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/  
>>> Last update of whois database: 2025-11-23T13:07:53Z <<<  
  
For more information on Whois status codes, please visit https://icann.org/epp  
  
NOTICE: The expiration date displayed in this record is the date the  
registrar's sponsorship of the domain name registration in the registry is  
currently set to expire. This date does not necessarily reflect the expiration  
date of the domain name registrant's agreement with the sponsoring  
registrar. Users may consult the sponsoring registrar's Whois database to  
view the registrar's reported date of expiration for this registration.  
  
TERMS OF USE: You are not authorized to access or query our Whois  
database through the use of electronic processes that are high-volume and  
automated except as reasonably necessary to register domain names or  
modify existing registrations; the Data in VeriSign Global Registry  
Services' ("VeriSign") Whois database is provided by VeriSign for
```

 Domains Hosting Servers Email Security Whois De	
testfire.net	
Domain Information	
Domain:	testfire.net
Registered On:	1999-07-23
Expires On:	2026-07-23
Updated On:	2025-02-27
Status:	client delete prohibited client transfer prohibited client update prohibited
Name Servers:	asia3.akam.net eur2.akam.net eur5.akam.net ns1-206.akam.net ns1-99.akam.net usc2.akam.net usc3.akam.net usw2.akam.net

Information Collected:

1. Domain ownership and organization information
2. Registrar and registration authority
3. Name server records
4. Administrative and technical contact emails (if not redacted)
5. Domain creation date
6. Domain expiration date
7. Domain update/modification records
8. DNSSEC status

2.4 FINDINGS SUMMARY

1. Domain Registration

- **Domain:** testfire.net
- **Registrar:** Amazon Registrar, Inc.
- **Creation Date:** 1999-07-23
- **Expiration Date:** 2026-07-23
- **Updated On:** 2025-02-27

2. WHOIS LOOKUP

2. Name Servers

- asia3.akam.net
- eur2.akam.net
- eur5.akam.net
- ns1-206.akam.net
- ns1-99.akam.net
- usc2.akam.net
- usc3.akam.net
- usw2.akam.net

3. Additional Information

- **Registrar abuse contact email:** trustandsafety@support.aws.com
- **Registrar IANA ID:** 468

2.5 IMPACT

WHOIS data provides crucial reconnaissance insights such as:

- Potential exposure of administrative emails
- Recon assistance for social engineering campaigns
- Understanding technology stack from name server providers
- Identifying domain ownership structure
- Supporting DNS-based enumeration (e.g., subdomains, MX records)

2.6 MITIGATIONS

To secure domain-related information:

- Enable **WHOIS privacy protection** if not already enabled
- Use generic administrative emails instead of personal ones
- Implement strict DNS configuration and monitoring
- Regularly review domain status to avoid expiration hijacking
- Ensure DNSSEC is enabled (if supported)

2.7 CONCLUSION

WHOIS lookup is an essential passive reconnaissance technique that reveals important domain details without sending direct requests to the target's infrastructure. The data collected for **testfire.net** helps build an understanding of the domain's background, hosting infrastructure, and potential areas that could be leveraged during further penetration testing.

3. IP ADDRESS LOOKUP

3. IP Address Lookup

IP Address Lookup is a passive reconnaissance technique used to gather network-level details about a target domain. For **testfire.net**, identifying the underlying IP address and analyzing its associated information provides insights into hosting providers, geolocation, ASN (Autonomous System Number), and potential infrastructure exposure. This step helps map the target's external footprint before deeper testing begins.

3.1 OBJECTIVE

The objective of IP address lookup for **testfire.net** is to:

- Identify the IP address associated with the domain
- Gather geolocation information of the hosting server
- Determine the hosting provider or cloud infrastructure used
- Retrieve ASN information for network-level analysis
- Support future enumeration (e.g., port scanning, CDN detection)
- Provide context for risk evaluation

3.2 SCOPE

The scope includes all publicly available information obtained through IP and network lookup tools.

This assessment covers:

1. IP resolution for the domain
2. Geographical server location
3. ISP / Hosting provider details
4. ASN and routing information
5. Reverse DNS records

3.3 METHODOLOGY

Tools:

Online IP lookup service: <https://www.iplocation.net/>

3. IP ADDRESS LOOKUP

Fig. 3.3.1 Gathering information using iplocation.net

The figure shows the iplocation.net website interface on the left and a comparison of geolocation data from two services on the right.

iplocation.net Interface:

- Search bar with "Search" button.
- Navigation menu: MY IP, IP TRACKER, TOOLS, WEB, PRIVACY, CYBERSECURITY, API, GAME.
- Section: IP Address Lookup
- Text: Discover more
- Section: IP Location Finder
- Input field: 65.61.137.117
- Button: IP Lookup
- Text: Enter a IPv4, IPv6 or Domain name into the input box above, and we'll locate its IP location.

Geolocation data from IP2Location:

Geolocation data from IP2Location		Product: DB6, 2025-11-15
IP ADDRESS:	65.61.137.117	ISP: Rackspace Inc.
COUNTRY:	United States	ORGANIZATION: Not available
REGION:	Texas	LATITUDE: 29.4241
CITY:	San Antonio	LONGITUDE: -98.4937
Incorrect location?		Contact IP2Location
		view map

Geolocation data from ipinfo.io:

Geolocation data from ipinfo.io		Product: API, real-time
IP ADDRESS:	65.61.137.117	ISP: Not available
COUNTRY:	United States	ORGANIZATION: AS33070 Rackspace Hosting
REGION:	Texas	LATITUDE: 32.7831
CITY:	Dallas	LONGITUDE: -96.8067
Incorrect location?		Contact ipinfo.io
		view map

3.4 FINDINGS SUMMARY

IP Resolution:

- **Domain:** testfire.net
- Resolved IP Address: **65.61.137.117**

Hosting Provider:

- Country/region: San Antonio, Texas, United States
- Organization: AS33070 Rackspace Hosting

3.5 IMPACT

IP analysis helps determine:

- Whether the server is shared or dedicated
- The hosting environment where the application resides

Although IP data is public, attackers often use it to:

- Perform targeted port scanning
- Map network infrastructure
- Attempt geo-based attack

3.6 RECOMMENDATIONS

To improve security posture:

- Use a secure and reputable hosting provider
- Ensure that the IP is properly hardened and monitored

3. IP ADDRESS LOOKUP

3.7 CONCLUSION

IP Address Lookup is a crucial passive reconnaissance step that reveals important network-level information about **testfire.net**. Understanding the resolved IP, its hosting environment, and routing provides the foundation for deeper security testing such as port scanning, service enumeration, and exploit surface evaluation.

4. DNS FOOTPRINTING

4. DNS Footprinting

DNS Footprinting is a passive enumeration technique used to gather detailed information about a target's DNS infrastructure. Tools like **NSlookup** and **DIG** allow penetration testers to retrieve DNS records, analyze domain configurations, and identify potential misconfigurations or attack vectors. For this assessment, DNS footprinting was conducted to understand the domain's DNS structure, hosting environment, and publicly exposed information.

4.1 OBJECTIVE

The main objectives of DNS footprinting using **NSlookup** and **DIG** are:

- Identify DNS servers (authoritative & non-authoritative)
- Enumerate DNS records (A, MX, NS, TXT, SOA, CNAME, etc.)
- Understand domain hosting architecture
- Investigate DNS misconfigurations
- Collect information useful for further reconnaissance steps
- Validate whether DNS records expose sensitive internal data

4.2 SCOPE

Scope includes:

- DNS server enumeration
- DNS record analysis
- Zone transfer checks (AXFR)
- Subdomain discovery (via known records)
- Email infrastructure identification (MX records)

4.3 METHODOLOGY

Tools Used:

- **NSlookup** (Windows, Linux, macOS)
- **DIG** (Linux, macOS, or Windows via BIND utilities)

Steps Performed:

1. Using NSlookup:

- Queried the domain to identify default DNS server
- Retrieved A record (IPv4 address)
- Retrieved NS records for authoritative name servers
- Checked MX and TXT records

4. DNS FOOTPRINTING

- Tested for open zone transfers

fig. 4.3.1 Queried the Domain to Identify Default DNS Server

```
(sai@kali)-[~]
$ nslookup testfire.net
Server:      192.168.222.243
Address:     192.168.222.243#53

Non-authoritative answer:
Name:   testfire.net
Address: 65.61.137.117
```

`nslookup zonetransfer.me`

fig. 4.3.2 Retrieved a Record (IPV4 Address)

```
(sai@kali)-[~]
$ nslookup -type=A zonetransfer.me
Server:      192.168.222.243
Address:     192.168.222.243#53

Non-authoritative answer:
Name:   zonetransfer.me
Address: 5.196.105.14
```

`nslookup -type=A zonetransfer.me` (For IPV4)

fig. 4.3.3 Retrieved NS Records

```
(sai@kali)-[~]
$ nslookup -type=NS zonetransfer.me
Server:      192.168.222.243
Address:     192.168.222.243#53

Non-authoritative answer:
zonetransfer.me nameserver = nsztl2.digi.ninja.
zonetransfer.me nameserver = nsztl1.digi.ninja.

Authoritative answers can be found from:
```

`nslookup -type=NS zonetransfer.me` (For Name Server)

4. DNS FOOTPRINTING

fig. 4.3.4 Checked for MX Records SOA Records

```
(sai@kali)-[~]
$ nslookup -type=MX zonetransfer.me
Server:      192.168.222.243
Address:     192.168.222.243#53

Non-authoritative answer:
zonetransfer.me mail exchanger = 10 ALT1.ASPMX.L.GOOGLE.COM.
zonetransfer.me mail exchanger = 20 ASPMX3.GOOGLEMAIL.COM.
zonetransfer.me mail exchanger = 20 ASPMX4.GOOGLEMAIL.COM.
zonetransfer.me mail exchanger = 0 ASPMX.L.GOOGLE.COM.
zonetransfer.me mail exchanger = 20 ASPMX2.GOOGLEMAIL.COM.
zonetransfer.me mail exchanger = 20 ASPMX5.GOOGLEMAIL.COM.
zonetransfer.me mail exchanger = 10 ALT2.ASPMX.L.GOOGLE.COM.

Authoritative answers can be found from:
zonetransfer.me nameserver = nsztm1.digi.ninja.
zonetransfer.me nameserver = nsztm2.digi.ninja.

(sai@kali)-[~]
$ nslookup -type=SOA zonetransfer.me
Server:      192.168.222.243
Address:     192.168.222.243#53

Non-authoritative answer:
zonetransfer.me
  origin = nsztm1.digi.ninja
  mail addr = robin.digi.ninja
  serial = 2019100801
  refresh = 172800
  retry = 900
  expire = 1209600
  minimum = 3600

Authoritative answers can be found from:
zonetransfer.me nameserver = nsztm1.digi.ninja.
zonetransfer.me nameserver = nsztm2.digi.ninja.
```

`nslookup -type=MX zonetransfer.me` (Mail Servers)

`nslookup -type=SOA zonetransfer.me`

2. Using DIG

- Performed record enumeration (A, AAAA, CNAME, MX, NS, TXT, SOA)
- Investigated DNS propagation
- Checked for possible misconfigurations
- Tested AXFR zone transfer using:
dig domain.com AXFR @nameserver

4. DNS FOOTPRINTING

fig. 4.3.5 Performed Record enumeration for MX

```
(sai@kali)-[~]
$ dig zonetransfer.me MX

; <<>> DiG 9.20.11-4+b1-Debian <<>> zonetransfer.me MX
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 14157
;; flags: qr rd ra; QUERY: 1, ANSWER: 7, AUTHORITY: 2, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 1280
;; QUESTION SECTION:
;zonetransfer.me.                IN      MX

;; ANSWER SECTION:
zonetransfer.me.                7200    IN      MX      10 ALT1.ASPMX.L.GOOGLE.COM.
zonetransfer.me.                7200    IN      MX      20 ASPMX3.GOOGLEMAIL.COM.
zonetransfer.me.                7200    IN      MX      10 ALT2.ASPMX.L.GOOGLE.COM.
zonetransfer.me.                7200    IN      MX      20 ASPMX5.GOOGLEMAIL.COM.
zonetransfer.me.                7200    IN      MX      0  ASPMX.L.GOOGLE.COM.
zonetransfer.me.                7200    IN      MX      20 ASPMX2.GOOGLEMAIL.COM.
zonetransfer.me.                7200    IN      MX      20 ASPMX4.GOOGLEMAIL.COM.

;; AUTHORITY SECTION:
zonetransfer.me.                7163    IN      NS       nsztm2.digi.ninja.
zonetransfer.me.                7163    IN      NS       nsztm1.digi.ninja.

;; Query time: 388 msec
;; SERVER: 192.168.222.243#53(192.168.222.243) (UDP)
```

dig zonetransfer.me MX

fig. 4.3.6 Performed Record Enumeration For TXT

```
(sai@kali)-[~]
$ dig zonetransfer.me TXT

; <<>> DiG 9.20.11-4+b1-Debian <<>> zonetransfer.me TXT
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 55736
;; flags: qr rd ra; QUERY: 1, ANSWER: 1, AUTHORITY: 2, ADDITIONAL: 1

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags;; udp: 1280
;; QUESTION SECTION:
;zonetransfer.me.                IN      TXT

;; ANSWER SECTION:
zonetransfer.me.                301     IN      TXT      "google-site-verification=typ28J7JAUHA9fw2sHXMGcCC0I6XBmmoVi04VLMewxA"

;; AUTHORITY SECTION:
zonetransfer.me.                7097    IN      NS       nsztm2.digi.ninja.
zonetransfer.me.                7097    IN      NS       nsztm1.digi.ninja.

;; Query time: 372 msec
;; SERVER: 192.168.222.243#53(192.168.222.243) (UDP)
```

dig zonetransfer.me TXT

4. DNS FOOTPRINTING

fig. 4.3.7 Tested AXFR Transfer

```
[sasi@kali:~]-[+]
; dig axfr zonetransfer.me @nsztmt.digi.ninjaja

;<>> Dig 9.20.11+4bl-Debian <>> axfr zonetransfer.me @nsztmt.digi.ninjaja
;; global options: +cmd
zonetransfer.me.      7200    IN      SOA     nsztmt.digi.ninja. robin.digi.ninja. 2019100801 172800 900 1209600 3
600
zonetransfer.me.      7200    IN      MXSKE   256 3 7 AWAAApal-InQByx2o1sdIz424-eDFgFnVWBCOfNFVCy3LrngZsBsEurB Byh
MQOQSIOYvU/7b3cBTj2BWJqGuxZe79QHWS7BFK7D-bP/AosnB05 KsgJKXvpHtJ9-B/XBV9TJAwL126fDA4AxsrspFbzr29SVDPM WBg/LM
aqkTM9HyvP+Nuzr0EPNhoXs/iPM+SqrVT8Fr6ydw2YPtyVDW iIkN760QBshXkjdlTyTQQiohkqzbypBJROJ37K3NLdCBoaoZoXhs/Rfl DQzxzy
YSVbfLwmlueuo7YA1lI/AHQ7DSUsFu2SzvGCyRbmmygyBN vABSTVF2ISem
zonetransfer.me.      301    IN      TXT     google-site-verification=tp28373AUHAFA5w2HXmGCCccIT8EXmmoVi04VLMEwx
A
zonetransfer.me.      7200    IN      MX      @ ASPMX.L.GOOGLE.COM.
zonetransfer.me.      7200    IN      MX      10 ALT1.ASPMX.L.GOOGLE.COM.
zonetransfer.me.      7200    IN      MX      10 ALT2.ASPMX.L.GOOGLE.COM.
zonetransfer.me.      7200    IN      MX      20 ASPMX2.GOOGLMAIL.COM.
zonetransfer.me.      7200    IN      MX      20 ASPMX3.GOOGLMAIL.COM.
zonetransfer.me.      7200    IN      MX      20 ASPMX4.GOOGLMAIL.COM.
zonetransfer.me.      7200    IN      MX      20 ASPMX5.GOOGLMAIL.COM.
zonetransfer.me.      7200    IN      A       5.196.105.14
zonetransfer.me.      7200    IN      NS       nsztmt.digi.ninja.
zonetransfer.me.      7200    IN      NS       nsztmt2.digi.ninja.
zonetransfer.me.      7200    IN      CERT    PMIX 0 0 fllTDVTCaJCFrhSBSGcpVyxzhsgdImaDUeUSMAACScGSib3DEE Cw
UAMIGAKswxCQVQGEJHQJfYMVGAlUECWAmPenpuZJRKYMSZmwY LmlMRQGWyJkoZHvcNAQKRfsSdg6JAGIeVa5DuawSqVTAFeWnyNTAJ MDIMXe
UAAGGAEGUCWNMFSGfaIUZZIEYMVGAlUECWAmPenpuZJRKYMSZmwY LmlMRQGWyJkoZHvcNAQKRfsSdg6JAGIeVa5DuawSqVTAFeWnyNTAJ MDIMXe
UIMTNaFWbyNJIA3DMTXUMTNaMIgaMcGUQCVDDGGWMyJa3MOJEY MBYGAlUECWAP29IdggGWWyJa3MOJEYMBYGADYDVGHQDAITAgnvZnl1 bGQXEIAQ
BGNVBAOMCUR22LIuasSYTEOMAAGAUGOCWPSGFIZUZIEY MBYGAlUECWAmPenpuZJRKYMSZmwY LmlMRQGWyJkoZHvcNAQKRfsS dgAIeZlna5
ecgzljYFUUME0NOxytkuPAfPdzFoTVotUDYFCVMYvrEQ Zz4kuORIEXPBMHWZQSMakwVLLIB/FES96fegvgXgw533QatKAATVCH DKTLfwvvh/cC-
0GIEmbwGduZFMmf/SymmkGDxEWB7q3+SBYfvRUohC HJGVNIEdwk6veUBTIj+3+i4SZKLXIzfEFfcIK/MQOVdfwpbjpw+cWGTo UBvfQZFzaPt9+9V
Br0f8ldrfgfj/MaeInrsHeuXotskd3PAQAQWGXz N3UVINEBOAACWEAAETAAnbgkhqkI09wBBAGAOEAQOXpfptvz/f7 tkkgfK27BdiZlu
Wkb08yRkVRAtAwMKwDb9ZdvvtQIB XGBILQXGvGvYfW3+2eesNgvrvIVxbULVicEt DgmImkrZewuluNoqaqd
TW3DG24ECMPQPKVPFH9u3jPaF23SLip SdpujpBKslfYruZJVwwcyTKUKMas115v78B9WM31EVL2PBIC AdWFylfx-tuteMe6Izsxxkl
tp0/Etlb39cnKgFGtsRD23jyb3CCily m4ANUZOmdNCACOT7uzXbeZ6lIFSNagQvyFYomFPnaheCYRHWE=
zonetransfer.me.      300    IN      HINFO   "Casio fx-7000" Windows XP
_acme-challenge.zonetransfer.me. 301 IN TXT "60a8ShBUJ9svvY7pAQvwCU5SGxvrbdizjePesZI"
cip.to.zonetransfer.me. 14000 IN SRV _sip._tcp.zonetransfer.me. 14000 IN SRV
I4.105.196.5-IN-ADDR.ARPA.zonetransfer.me. 7200 IN PTR www.zonetransfer.me.
asfdbauthdns.zonetransfer.me. 7900 IN AFSDDB 1 asfdbbox.zonetransfer.me.
asfdbbox.zonetransfer.me. 7200 IN A 127.0.0.1
asfdbnull.zonetransfer.me. 7200 IN AFSDDB 1 asfdbbox.zonetransfer.me.
canberra-office.zonetransfer.me. 7200 IN A 202.14.81.230
cmdeexc.zonetransfer.me. 7200 IN TXT "; ls"
contact.zonetransfer.me. 2592000 IN TXT "Remember to call Pippa on +44 123 4567890 or pippa@zonetra
nsofer.me when making DNS changes"
dc-office.zonetransfer.me. 7200 IN A 143.228.181.132
deadbeef.zonetransfer.me. 7201 IN AAAA dead:beaf::
dr.zonetransfer.me. 7200 IN LOC 53 20 56.958 N 1 38 33.526 W 0.00m 1m 10000m 10m
DCZ.zonetransfer.me. 7200 IN TXT "AbcdEfG"
email.zonetransfer.me. 7222 IN NAPTR ".*?P.*?" "E2U+mail" "" email.zonetransfer.me.zonetransfer.me.
email.zonetransfer.me. 7200 IN A 74.125.206.26
Hello.zonetransfer.me. 7200 IN TXT "Hi to Josh and all his class"
home.zonetransfer.me. 7200 IN A 127.0.0.1
info.zonetransfer.me. 7200 IN A info.zonetransfer.me service provided by Robin Wood - robing@digi.ninja.
See http://ddgi.ninja/projects/zonetransferme.php for more information."
internal.zonetransfer.me. 300 IN NS intns1.zonetransfer.me.
internal.zonetransfer.me. 300 IN NS intns2.zonetransfer.me.
intns1.zonetransfer.me. 300 IN A 81.4.108.41
intns2.zonetransfer.me. 300 IN A 5.196.105.14
office.zonetransfer.me. 7200 IN A 4.23.39.254
ip6vactnow.org.zonetransfer.me. 7200 IN AAAA 2001:67c:2eb:f11:c100:1332
owa.zonetransfer.me. 7200 IN A 207.46.197.32
robinwood.zonetransfer.me. 302 IN TXT "Robin Wood"
rp.zonetransfer.me. 321 IN RP robin.zonetransfer.me. robinwood.zonetransfer.me.
sip.zonetransfer.me. 3333 IN NAPTR "2.*?P.*?" "E2U+sip" "f..&.sip:customer-service@zonetransfer.me" ".
sql.zonetransfer.me. 300 IN TXT "or ip1 .."
zonetransfer.me. 7200 IN TXT "(.) (.*) echo Shell Shocked"
staging.zonetransfer.me. 7200 IN CNAME www.gdyneyoperahouse.com.
allccportsopen.firewall.test.zonetransfer.me. 301 IN A 127.0.0.1
testing.zonetransfer.me. 301 IN CNAME www.zonetransfer.me.
vpn.zonetransfer.me. 4000 IN A 174.36.59.154
zonetransfer.me. 7200 IN A 5.196.105.14
xss.zonetransfer.me. 7200 IN TXT "<script>alert('boo')</script>"
zonetransfer.me. 7200 IN SOA nsztmt.digi.ninja. robin.digi.ninja. 2019100801 172800 900 1299600 3
600
Query time: 264 msec
SERVER: 81.4.108.41(53(nsztmt.digi.ninja) (TCP)
WHEN: Mon Nov 24 08:44:54 IST 2025
XFR size: 52 records (messages 1, bytes 3339)
```

dig axfr zonetransfer.me @nsztm1.digi.ninja

Information Collected:

- Domain-to-IP mapping
- Name servers and DNS provider
- Email routing server info
- Additional hostnames from DNS records
- DNS TTL values and configuration metadata

4. DNS FOOTPRINTING

4.4 FINDINGS SUMMARY

DNS Servers Identified

- Nsztm1.digi.ninja
- Nsztm2.digi.ninja

DNS Records Found

- List of NS servers
- MX records
- A/AAAA records
- TXT records (SPF, test entries)
- SOA information (primary DNS, admin email)

Zone Transfer Test

The successful zone transfer exposed:

- Full list of subdomains
- Internal service names
- Development and staging hostnames
- Email routing information
- Potential attack surface mapping

Potential Risks Identified

- Sensitive DNS information exposure
- Increased attack surface due to visible internal hosts
- Potential for social engineering based on discovered hostnames
- Support for targeted reconnaissance (e.g., subdomain scanning)

4.5 IMPACT

- Map the entire network structure
- Identify internal hosts and services
- Target exposed subdomains
- Perform focused attacks such as phishing or brute-force
- Bypass security by exploiting DNS misconfigurations

Such exposure significantly increases the attack surface and is considered a serious DNS misconfiguration.

4.6 MITIGATIONS

To secure DNS configurations:

- Disable zone transfer to unauthorized hosts

4. DNS FOOTPRINTING

- Restrict AXFR to specific trusted IP addresses
- Use DNSSEC where possible
- Regularly audit DNS configurations
- Avoid exposing unnecessary DNS records
- Monitor DNS queries for suspicious behavior

4.7 CONCLUSION

DNS footprinting using NS Lookup and Dig provides valuable insight into a domain's network infrastructure. Zone transfer testing is an essential check to ensure DNS servers are configured securely. The controlled domain **zonetransfer.me** successfully demonstrated how misconfigured DNS servers can leak sensitive information, highlighting the importance of proper DNS security measures.

4.8 MORE ALTERNATIVES

1. DNSRecon:

Fig. 4.8.1 Performing DNS Enumeration using DNSRecon

```
(sai@kali)-[~]
$ dnsrecon -d zonetransfer.me -k
[*] std: Performing General Enumeration against: zonetransfer.me...
[*] DNSSEC is configured for zonetransfer.me
[*] DNSKEYs:
[*] None 75K RSASHA1NSEC3SHA1 03010001aa682f27401631da88b774 8e36e3e7440c5827556d1c38835f998d e32eb9e067106c12eafc07284c390b31 a0839
63def55da5b94b9ad27a878017b2 b6e04ebda0f4a3a23d221bbc33cc5bca 8b31a4244e6f3d86f3dcf8dbb3ace10e 3cd8685ecfe23ccf92433aef4c17ebbb 2d30db23e
cf1cd7c98495f1f7f8a651eb a5a3b600d7523f00743b0ec52c16e092 dafca60b247c9ebc7d8180cde2c06f4c 5da2e5e3
[*] SOA nsztm1.digi.ninja 81.4.108.41
[*] NS nsztm1.digi.ninja 81.4.108.41
[*] Bind Version for 81.4.108.41 secret*
[*] NS nsztm2.digi.ninja 5.196.105.10
[*] Bind Version for 5.196.105.10 secret*
[*] MX ASPMX3.GOOGLEMAIL.COM 172.217.78.27
[*] MX ASPMX4.GOOGLEMAIL.COM 142.250.101.26
[*] MX ALT2.ASPMX.L.GOOGLE.COM 172.217.78.26
[*] MX ALT3.ASPMX.L.GOOGLE.COM 192.178.163.27
[*] MX ASPMX2.GOOGLEMAIL.COM 192.178.163.26
[*] MX ASPMX1.L.GOOGLE.COM 64.233.178.27
[*] MX ASPMX5.GOOGLEMAIL.COM 192.178.164.27
[*] MX ASPMX3.GOOGLEMAIL.COM 2607:f8b0:4023:1c05::1a
[*] MX ASPMX4.GOOGLEMAIL.COM 2607:f8b0:4023:c06::1b
[*] MX ALT2.ASPMX.L.GOOGLE.COM 2607:f8b0:4023:1c05::1a
[*] MX ALT3.ASPMX.L.GOOGLE.COM 2607:f8b0:400e:c17::1b
[*] MX ASPMX2.GOOGLEMAIL.COM 2607:f8b0:400e:c17::1a
[*] MX ASPMX.L.GOOGLE.COM 2404:6800:4003:c03::1b
[*] MX ASPMX5.GOOGLEMAIL.COM 2607:f8b0:4023:2009::1b
[*] A zonetransfer.me 5.196.105.14
[*] TXT zonetransfer.me google-site-verification=tyP28J7JAUHA9fwzSHXMGccC0I6XBmoV104VIMewxA
[*] Enumerating SRV Records
[*] SRV _sip._tcp.zonetransfer.me www.zonetransfer.me 5.196.105.14 5060
[*] 1 Records Found
[*] Performing crt.sh Search Enumeration
```

dnsrecon -d zonetransfer.me -k

2. Amass:

fig. 4.8.2 Performing DNS Enumeration using Amass

```
(sai@kali)-[~]
$ amass enum -d zonetransfer.me
zonetransfer.me (FQDN) → mx_record → alt1.aspmx.l.google.com (FQDN)
zonetransfer.me (FQDN) → mx_record → aspmx.l.google.com (FQDN)
zonetransfer.me (FQDN) → mx_record → aspmx4.googlemail.com (FQDN)
zonetransfer.me (FQDN) → mx_record → aspmx5.googlemail.com (FQDN)
zonetransfer.me (FQDN) → mx_record → aspmx3.googlemail.com (FQDN)
zonetransfer.me (FQDN) → mx_record → alt2.aspmx.l.google.com (FQDN)
zonetransfer.me (FQDN) → mx_record → aspmx2.googlemail.com (FQDN)

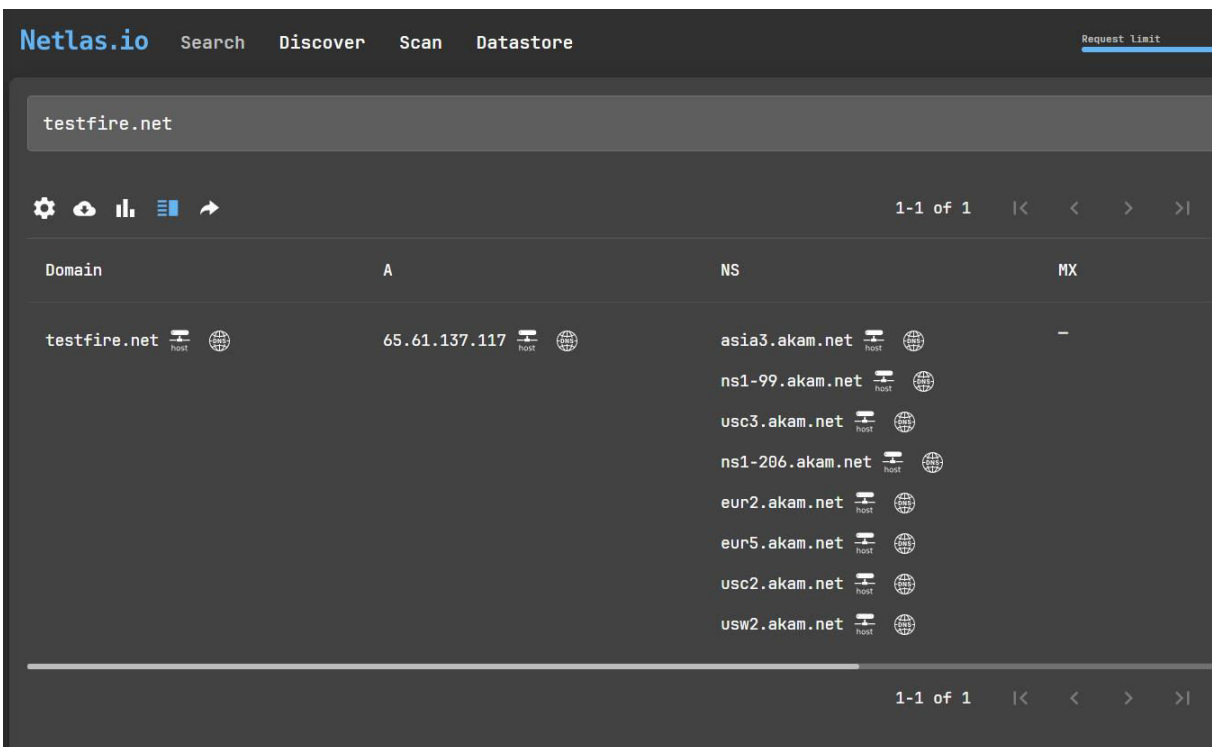
The enumeration has finished
```

amass enum -d zonetransfer.me

4. DNS FOOTPRINTING

3. Nelas.io

Fig. 4.8.3 DNS Footprinting using netlas.io



The screenshot shows the Netlas.io web application interface. At the top, there is a navigation bar with the logo 'Netlas.io' and links for 'Search', 'Discover', 'Scan', and 'Datastore'. A 'Request Limit' indicator is on the right. Below the navigation bar, a search bar contains the text 'testfire.net'. Underneath the search bar, there are several icons (gear, cloud, bar chart, list, and share) and a pagination indicator '1-1 of 1'. The main content area displays a table with DNS records for 'testfire.net'. The table has four columns: 'Domain', 'A', 'NS', and 'MX'. The 'Domain' column shows 'testfire.net' with host and globe icons. The 'A' column shows the IP address '65.61.137.117' with host and globe icons. The 'NS' column lists eight nameservers: 'asia3.akam.net', 'ns1-99.akam.net', 'usc3.akam.net', 'ns1-206.akam.net', 'eur2.akam.net', 'eur5.akam.net', 'usc2.akam.net', and 'usw2.akam.net', each with host and globe icons. The 'MX' column shows a hyphen '-'. At the bottom right, there is another pagination indicator '1-1 of 1'.

Domain	A	NS	MX
testfire.net  	65.61.137.117  	asia3.akam.net   ns1-99.akam.net   usc3.akam.net   ns1-206.akam.net   eur2.akam.net   eur5.akam.net   usc2.akam.net   usw2.akam.net  	-

5. EMAIL FOOTPRINTING

5. Email Footprinting

Email footprinting is a passive reconnaissance technique used to identify publicly available email addresses associated with a target organization. For this assessment, the **GSA Email Spider** tool was used to scan publicly indexed web pages for email IDs related to **testfire.net**.

This information helps understand potential exposure to phishing, social engineering, and spam-based attacks.

5.1 OBJECTIVE

- To identify any **publicly exposed email addresses** associated with the domain.
- To assess the scope of **email leakage** across the web.
- To detect **employee/department emails** that could be used in social engineering.
- To evaluate whether email harvesting protection mechanisms exist.
- To support further phases of security testing (OSINT, phishing simulation, etc.).

5.2 SCOPE

- Websites indexed by search engines
- Public web pages referencing testfire.net
- Social media or open forums where the domain appears
- Email ID metadata that the GSA tool can harvest

5.3 METHODOLOGY

1. Tool Used

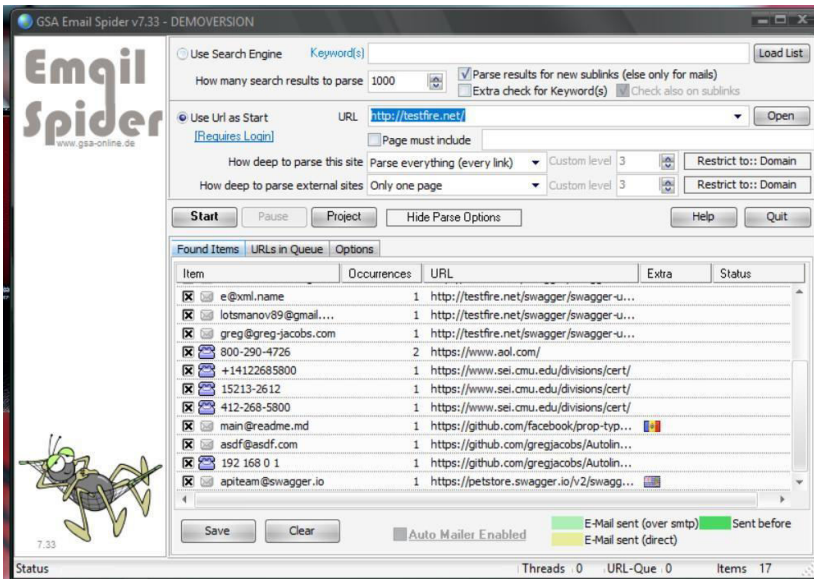
- **GSA Email Spider**
Used to automatically crawl web pages, extract email IDs, and identify patterns related to the target domain.

2. process

1. Configured GSA Email Spider to target **testfire.net** specifically.
2. Crawled:
 - Public websites
 - Search engine results
 - Social sources

5. EMAIL FOOTPRINTING

fig. 5.3.1 GSA Email Spyder



3. Filtered extracted data to include only:
 - @testfire.net addresses
 - Emails referencing TestFire, Altoro Mutual, or related names
4. Validated email formats using pattern matching.
5. Compiled and categorized findings.

5.4 IMPACT

Exposure of email addresses may lead to:

- **Phishing attacks** targeting employees
- Credential harvesting attempts
- **Social engineering** using impersonation
- Spam or malicious email flooding
- Attackers mapping organization hierarchy using email naming patterns

5.5 RECOMMENDATIONS

- Use **generic emails** (e.g., info@domain.com) instead of personal emails on public pages.
- Implement **email obfuscation techniques**:
 - JavaScript-based email rendering
 - Image-based email display
 - Contact forms instead of direct email links
- Regularly scan the internet for leaked email addresses.

5. EMAIL FOOTPRINTING

- Educate employees about phishing and impersonation threats.
- Enabled advanced spam filters and DMARC/DKIM/SPF policies.

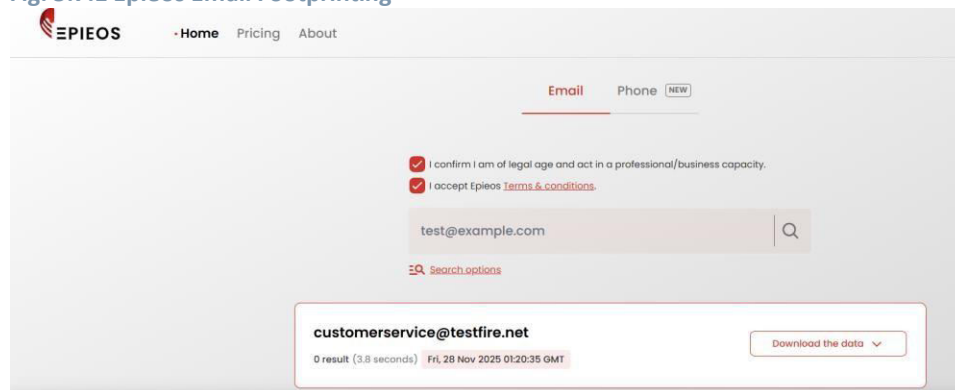
5.6 CONCLUSION

Email Footprinting using **GSA Email Spider** provides insight into publicly accessible email data associated with **testfire.net**. This passive reconnaissance step helps understand an organization's exposure to phishing and social engineering attacks. The results assist in strengthening email privacy and improving security awareness.

5.7 OTHER RELATED TOOLS

- 1) **Epieos**: Lets you enter an email and (on free tier) checks across many public services to see where that email is used — linked Google accounts, social-profiles, past registrations, etc.

Fig. 5.7.1 Epieos Email Footprinting



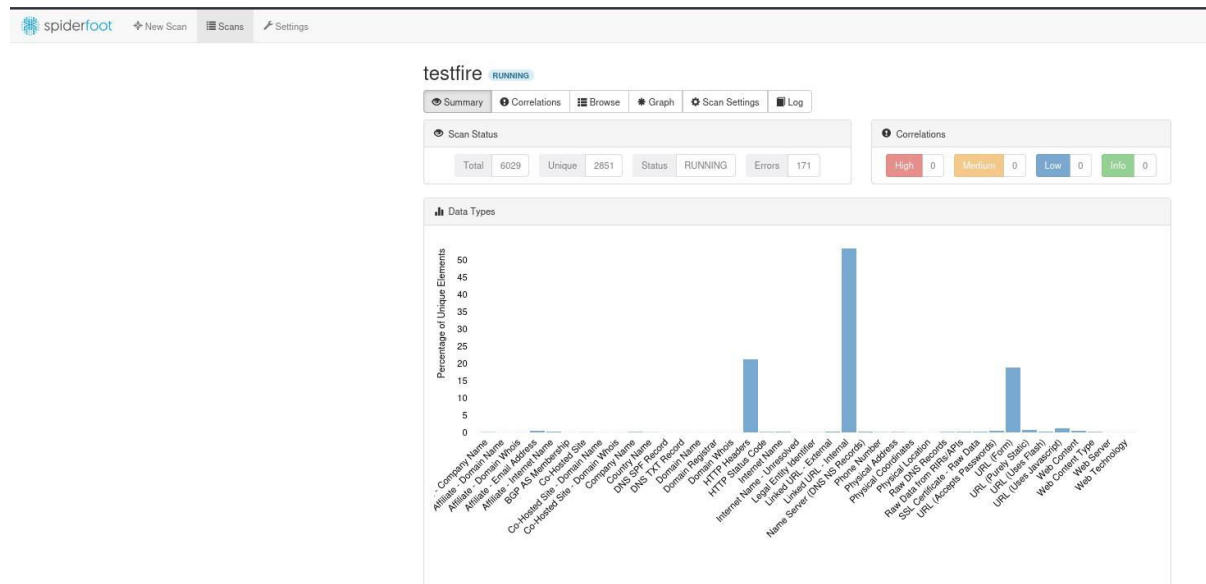
- 2) **Spiderfoot**: Open-source / self-hostable OSINT automation tool — can scan a target email (or domain) across many data sources (social media, breaches, associated domains, IPs, metadata) and help build a broader footprint.

To start spiderfoot in kali use command:

- i) `spiderfoot -l 127.0.0.1:5001`
- ii) Enter the Domain name testfire.net at Scan Target.
- iii) Name the Scan
- iv) Select scan Type By Use case
- v) Then Click on Run Scan

5. EMAIL FOOTPRINTING

Fig. 5.7.2 Spiderfoot Footprinting



- 3) **theHarvester**: Popular OSINT reconnaissance tool: can harvest emails, subdomains, names, and other public data — often used in initial reconnaissance to collect whatever is publicly available.
- i) Use command to run the tool:
`theharvester -d testfire.net -b all`

fig. 5.7.3 Using theHarvester For Footprinting

```
[*] LinkedIn Links found: 0

[*] IPs found: 1
65.61.137.117

[*] No emails found.

[*] No people found.

[*] Hosts found: 50
28demo.testfire.net
3ddemo.testfire.net
Demo.testfire.net
altoro.testfire.net
altoro.testfire.net:65.61.137.117
altoroc.testfire.net
altoro--.testfire.net
demo-analytics.testfire.net
demo-analyticस्क.testfire.net
demo.testfire.net
demo.testfire.net.Demo.testfire.net
demo.testfire.net:64:ff9b::413d:8975
demo.testfire.net:65.61.137.117
demo2.testfire.net
demo2.testfire.net:65.61.137.117
demo2--.testfire.net
demo--.testfire.net
evil.testfire.net:64:ff9b::413d:8975
evil.testfire.net:65.61.137.117
evil--.testfire.net
ftp.testfire.net
ftp.testfire.net:testfire.net
ftp--.testfire.net
```

5. EMAIL FOOTPRINTING

- 4) **h8mail**: CLI / OSINT tool (open-source) that helps check if an email appears in public breaches and leak databases — useful for assessing whether email was compromised or exposure exist
- a) use command to run the tool
- ```
h8mail -t admin@testfire.net
```

fig. 5.7.4 using tool h8mail for email Footprinting

```

(sai@kali)~$ h8mail -t admin@testfire.net
Official h8mail posts:
https://khast3x.club/tags/h8mail/

Version 2.5.6 - "ROCKSMASION.6"

 h8mail
 Heartfelt Email OSINT
 Use responsibly
 github.com/khast3x

[>] h8mail is up to date
[~] Removing duplicates
[>] Targets:
[>] admin@testfire.net
[~] scylla.so is down, skipping
[~] Target factory started for admin@testfire.net
[~] [admin@testfire.net]>[hunter.io public]
[!] hunter.io (public API) error: admin@testfire.net
'data'

[>] Showing results for admin@testfire.net
[~] No results founds

Session Recap:

Target	Status
 admin@testfire.net | Not Compromised

Execution time (seconds): 7.6812074184417725

```

## 6. NETWORK FOOTPRINTING

### 6. Network Footprinting

Network footprinting is a critical phase during the reconnaissance stage of a penetration test. Using tools like **Angry IP Scanner**, security testers can identify live hosts, open ports, and basic network characteristics across an IP range. This technique provides a quick overview of the network's active devices and their publicly exposed services.

#### 6.1 OBJECTIVE

The primary objectives of performing network footprinting with Scanner are:

- Identify active/live hosts within the target IP range
- Enumerate open ports and basic host details
- Detect devices or services unintentionally exposed
- Understand the external network structure and possible entry points
- Provide a foundation for deeper scanning (Nmap, Nessus, etc.)

#### 6.2 SCOPE

The scope of this footprinting activity includes:

- Scanning a defined range of IP addresses
- Identifying alive hosts
- Detecting open ports (Ping scan + TCP port scan)
- Collecting hostname and response-time data

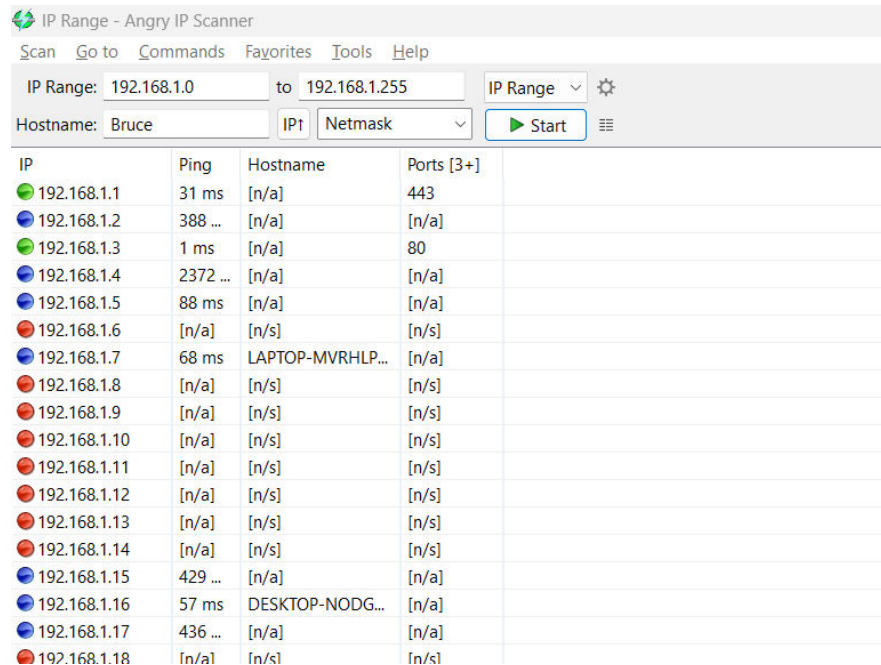
#### 6.3 METHODOLOGY

- 1) **Tools Used:** Angry IP Scanner
- 2) **Steps Performed:**
  - a) Defined the target IP range related to the assessment scope
  - b) Configured scan options:
    - i) Ping scan to identify live hosts
    - ii) Basic port scan on common ports
    - iii) Hostname detection
  - c) Analyzed the results for:
    - i) Active devices

## 6. NETWORK FOOTPRINTING

- ii) Open ports
- iii) Hostname or domain mappings
- iv) Response times
- d) Documented identified hosts and services for further investigation

fig. 6.3.1 Network Scanning using Angry IP



| IP           | Ping     | Hostname         | Ports [3+] |
|--------------|----------|------------------|------------|
| 192.168.1.1  | 31 ms    | [n/a]            | 443        |
| 192.168.1.2  | 388 ...  | [n/a]            | [n/a]      |
| 192.168.1.3  | 1 ms     | [n/a]            | 80         |
| 192.168.1.4  | 2372 ... | [n/a]            | [n/a]      |
| 192.168.1.5  | 88 ms    | [n/a]            | [n/a]      |
| 192.168.1.6  | [n/a]    | [n/s]            | [n/s]      |
| 192.168.1.7  | 68 ms    | LAPTOP-MVRHLP... | [n/a]      |
| 192.168.1.8  | [n/a]    | [n/s]            | [n/s]      |
| 192.168.1.9  | [n/a]    | [n/s]            | [n/s]      |
| 192.168.1.10 | [n/a]    | [n/s]            | [n/s]      |
| 192.168.1.11 | [n/a]    | [n/s]            | [n/s]      |
| 192.168.1.12 | [n/a]    | [n/s]            | [n/s]      |
| 192.168.1.13 | [n/a]    | [n/s]            | [n/s]      |
| 192.168.1.14 | [n/a]    | [n/s]            | [n/s]      |
| 192.168.1.15 | 429 ...  | [n/a]            | [n/a]      |
| 192.168.1.16 | 57 ms    | DESKTOP-NODG...  | [n/a]      |
| 192.168.1.17 | 436 ...  | [n/a]            | [n/a]      |
| 192.168.1.18 | [n/a]    | [n/s]            | [n/s]      |

### 6.4 FINDINGS SUMMARY

- 1) Active IP Addresses Identified:
  - o Number of live hosts: 105
  - o Hostnames resolved: LAPTOP-MVRHLP
  - o Notable devices discovered
- 2) Open Ports Detected
  - o Port 80 (HTTP) open on 192.168.1.3
  - o Port 443 (HTTPS) open on 192.168.1.1
  - o Port 22 (SSH) open on 192.168.1.78
- 3) Additional Observations
  - o High response-time hosts
  - o Unknown or suspicious services
  - o Hosts with multiple exposed services

## 6. NETWORK FOOTPRINTING

### 6.5 IMPACT

Network footprinting results can reveal:

- Misconfigured devices
- Unnecessary open ports
- Exposed services vulnerable to attack
- Devices not monitored or documented
- Lack of firewall or filtering controls

### 6.6 MITIGATIONS

To mitigate exposure and strengthen security posture:

- Restrict unnecessary services and close unused ports
- Implement strict firewall rules and network segmentation
- Disable external Ping responses if not required
- Monitor and log all inbound network traffic
- Ensure devices follow the principle of “least privilege”
- Regularly scan and audit IP ranges to detect rogue devices

### 6.7 CONCLUSION

Using Angry IP Scanner for network footprinting provides a fast, effective snapshot of active hosts and exposed network services. This baseline information is essential for identifying potential security weaknesses and preparing for deeper penetration testing phases.